BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final | Semester: Fall 2023
Duration: 2 hours | Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:

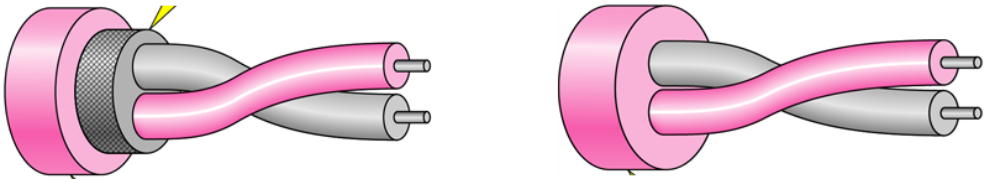
1. [CO3]

a) **Consider** there are five channels mentioned in the below table which are to be multiplexed in a way that the number of channels will **decrease**. If **one** synchronization bit is added to each frame and the interleaved unit from each channel is **2 bits**, **write** down the answers to the following questions. [2+8]

[***Hint: Convert the source speed to a common unit like Kbps (1000bits) or bps first and show the calculation for each question below***]

Channel Name	Data rate
A	10000 Character/s
B	10 KBytes/s
C	80 Kbps
D	5000 Character/s
E	5 KBytes/s

i. **Diagrammatically** represent the physical multiplexer with necessary details.
ii. In what kind of TDM, the channels are reduced?
iii. What is the **input bit** duration in **seconds**?
iv. What is the size of a frame in **bits**?
v. What is the frame rate (**FPS)**?
vi. What is the duration of a **frame** in **seconds**?
vii. What is the output data rate or transmission rate of the medium in **bps**?
viii. What is the output slot duration in **seconds**?
ix. How many overhead **bits** are passing through the medium per second? *[hint: frame rate]*

1. **[CO3]**	b)	**Sketch** the **Original Signal, Spread Code,** and **Spread Signal** from the following Original Signal (**010**) and the given Spreading Code (**10101**) using the concept of DSSS. You can use the **NRZ line coding scheme** (zero means low and one means high voltage) to draw the signals.	[3]
[CO3]	c)	What is the **drawback** of using DSSS?	[2]
2. **[CO4]**	a)	**Compare** the cable types in the following figure (mention two points). 	[2]
[CO4]	b)	**Identify** the transmission medium for the scenario given below: i. Suitable for moderate-speed networks such as cable TV and some broadband connections. ii. BNC connector is used. iii. Immune to electromagnetic interference.	[3]
3. **[CO5]**	a)	Assume a packet is made only of four **8-bit** words **$(100)_{10}$, $(20)_{10}$, $(221)_{10}$ and $(127)_{10}$** i. **Show** the checksum at the **sender**. ii. If the second data item is changed to **$(50)_{10}$** and the **third data** item is changed to **$(191)_{10}$** during transmission, check if any error can be detected at the **receiver site**. **Explain** your findings.	[5]

3. [CO5]	b)	Suppose two channels are using the following block coding scheme for error detection and correction. Dataword | Codeword 00 | 11110000 01 | 00001101 10 | 10111010 11 | 01110111 i. How many bits can it successfully **detect and correct** errors using this scheme? ii. The receiver received the codeword **10101010** that contains **1 bit error**. **Determine** if the original dataword can be retrieved from it or not?	[5]
[CO5]	c)	Assume the following cases for a sender and a receiver device that use CRC for error control ▪ Dataword: $x^6 + x^4 + x^3 + x + 1$ ▪ Divisor: $x^4 + x^2 + x + 1$ **Show** the **sent codeword** using CRC **polynomial** division.	[5]
4. [CO5]	a)	In CSMA/CD, what is done by a channel if a collision is detected during the transmission time? **Write** down the purpose of doing so.	[2]
[CO5]	b)	**Identify** the name of media access protocols for the scenario given below: i. Stations transmit data after receiving permission from a central controller. ii. Stations use different codes to achieve multiple access. iii. Stations take turns to transmit in a fixed order. iv. Stations contend for the channel and back off if the channel is busy. v. A station transmits its data in its assigned time slot only.	[5]
[CO5]	c)	For the 2nd attempt (k=2) in an **Ethernet-based MAC protocol**, what **are** the possible **Back-off times(T_B)** if the **Propagation time(T_P)** is 2 ms?	[3]
[CO5]	d)	In a CSMA/CD network running with a bandwidth of 1 Gbps over a cable, the signal speed in the cable is 20000 km/sec and the minimum frame size is 3000 bits. **Calculate** the **distance** covered by the signal.	[5]